

**Course Name: Computer Networks and Internet Protocol**

**Assignment - Week 12 (Jan 2026)**

**TYPE OF QUESTION: MCQ/MSQ**

**Number of questions: 10**

**Total mark: 10 X 1 = 10**

**QUESTION 1**

If there are  $n$  signal sources, each transmitting at the same data rate, how many time slots are required in a Time Division Multiplexing (TDM) link?

- a)  $n/2$
- b)  $2n$
- c)  $n$
- d)  $n^2$

**Correct Answer: (c)**

**Detailed Solution:** In TDM, the total time is divided into equal slots and each source is assigned one time slot. Therefore, for  $n$  sources, there are  $n$  time slots.

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**QUESTION 2**

Suppose we have a HDTV screen with screen resolution 1600 X 1200 and the screen is renewed 50 times per second. Considering a colored pixel needs 24 bits to be represented, what is the HDTV bit rate?

- a) 1.3 Gbps
- b) 2.3 Gbps
- c) 3.3 Gbps
- d) 4.3 Gbps

**Correct Answer: (b)**

**Detailed Solution:** Total pixels per frame = 1600 X 1200 . Bit per frame = 1600 X 1200 X 24. Bit rate = 1600 X 1200 X 24 X 50 = 2.3 Gbps

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**QUESTION 3**

A composite signal contains frequencies 300, 500, 800, 1000 and 1500 Hz. What is the bandwidth?

- a) 1000 Hz
- b) 200 Hz
- c) 1200 Hz
- d) 500 Hz

**Correct Answer: (c)**

**Detailed Solution:** Bandwidth = Highest frequency - Lowest frequency = 1500 - 300 = 1200 Hz.

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**QUESTION 4**

Which component is included in IP security?

- a) Authentication Header (AH)
- b) Encapsulating Security Payload (ESP)
- c) Internet key Exchange (IKE)
- d) All of the mentioned

**Correct Answer: (d)**

**Detailed Solution:** IPsec includes Authentication Header (AH) for integrity/authentication, Encapsulating Security Payload (ESP) for encryption and integrity, and Internet Key Exchange (IKE) for key management. Therefore, all listed components are part of IPsec.

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**QUESTION 5**

Which of the following is NOT a cause of transmission impairment?

- a) Attenuation
- b) Noise
- c) Distortion
- d) Congestion

**Correct Answer: (d)**

**Detailed Solution:** Transmission impairment refers to physical signal issues like attenuation, noise, and distortion. Congestion is a network level problem, not a signal impairment.

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### QUESTION 6

Which of the following is/are NOT true about modulation techniques?

- a) One amplitude in ASK is zero.
- b) BFSK is more susceptible to error than ASK.
- c) In Binary PSK, presence and absence of the carrier signal represents two binary digits.
- d) QAM is a combination of ASK and PSK

**Correct Answer : (b, c)**

**Detailed Solution:** In ASK, one amplitude can be zero (valid), and QAM is indeed a combination of amplitude and phase modulation (valid). However, BFSK is generally less susceptible to errors than ASK, so statement (b) is false. Also, in BPSK, data is represented by phase changes, not by presence or absence of a carrier, making (c) false.

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### QUESTION 7

Which of the following attacks can actively modify communications or data?

- a) Both active and passive attacks
- b) Neither active and passive attacks
- c) Active attacks
- d) Passive attacks

**Correct Answer: (c)**

**Detailed Solution:** Active attacks involve modification, insertion, or deletion of data, whereas passive attacks only monitor or eavesdrop without changing the data.

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### QUESTION 8

Which one is used in parts of the cellular telephone system and for some satellite communication?

- a) CDM
- b) TDM
- c) FDM
- d) WDM

**Correct Answer: (a)**

**Detailed Solution:** CDM (Code Division Multiplexing), specifically CDMA, is used in cellular systems and also in some satellite communications due to its ability to allow multiple users over the same frequency band.

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### QUESTION 9

Which of the following statements is/are correct?

- a) NAT is a proxy server
- b) NAT can also act as a firewall.
- c) NAT needs to recalculate both IP and TCP checksum as it modifies the IP header and port number.
- d) IP masquerading is done by NAT.

**Correct Answer: (b, c, d)**

**Detailed Solution:** NAT is not a proxy server, but it can act as a basic firewall by hiding internal addresses. Since NAT modifies IP addresses and port numbers, it must recalculate both IP and TCP checksums. It also performs IP masquerading.

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### QUESTION 10

If the frequency spectrum of a signal has a bandwidth of 500 Hz with the highest frequency at 600 Hz, what should be the sampling rate according to the Nyquist theorem?

- a) 200 samples/sec
- b) 500 samples/sec
- c) 1000 samples/sec
- d) 1200 samples/sec

**Correct Answer: (d)**

**Detailed Solution:** Bandwidth = 500 Hz, highest frequency = 600 Hz. Nyquist rate = 2 X highest frequency = 2 X 600 = 1200 samples/sec

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